

# Safety Data Sheet

According to Notification of Thailand's Ministry of Industry B.E. 2555 (2012)



## AKROS MULTI 10W30

Version 0.0

Release Date: 2017/01/25

### 1. PRODUCT AND COMPANY IDENTIFICATION

Product name : AKROS MULTI 10W30  
Other means of identification : None  
Recommended use : Lubricant oil  
Limitation of use : This product should not be used for purposes other than those specified without the advice of an expert  
Importer name : PETRONAS International Marketing (Thailand) Co.,Ltd  
Company registration number : 0105555014294  
Registered corporate address : 388 Exchange Tower, Unit Nos. 4101-4104 41<sup>st</sup> floor, Sukhumvit Road, Kweang Klongtoey, Khet Klongtoey, Bangkok 10110  
Emergency phone numbers : +662-663 5881-5

### 2. HAZARDS IDENTIFICATION

#### GHS Classification

Skin corrosion/irritation : Category 3  
Serious eye damage/eye irritation : Category 2A

#### GHS Label element

Hazard pictograms :



Signal word : Warning  
Hazard statements : H316 Causes mild skin irritation.  
H319 Causes serious eye irritation.  
Precautionary statements : **Prevention:**  
P264 Wash skin thoroughly after handling.  
P280 Wear eye protection/ face protection.  
**Response:**  
P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.  
P332 + P313 If skin irritation occurs: Get medical advice/ attention.  
P337 + P313 If eye irritation persists: Get medical advice/ attention.

#### Other hazards which do not result in classification

No information available.

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### 3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Mixture

#### Hazardous components

| Chemical Name                                             | CAS-No.      | Concentration (%) |
|-----------------------------------------------------------|--------------|-------------------|
| Paraffinic Distillate, Heavy, Hydrotreated                | 64742-54-7   | >= 70 - < 90      |
| Zinc Dialkyl Dithiophosphate                              | 68649-42-3   | >= 1 - < 3.5      |
| HOB Calcium Long Chain Alkaryl Sulfonate                  | Not Assigned | >= 1 - < 3.5      |
| Benzenesulfonic acid, propenated, calcium salt, overbased | 68610-84-4   | >= 0.1 - < 1      |

### 4. FIRST AID MEASURES

If inhaled : If unconscious, place in recovery position and seek medical advice.  
If symptoms persist, call a medical doctor.

In case of skin contact : If on skin, rinse well with water.  
If on clothes, remove clothes.  
If skin irritation persists, call a medical doctor.

In case of eye contact : Remove contact lenses.  
Protect unharmed eye.  
Keep eye wide open while rinsing.  
If eye irritation persists, consult a specialist.  
Immediately flush eye(s) with plenty of water.

If swallowed : Keep respiratory tract clear.  
Do NOT give milk or alcoholic beverages.  
Never give anything by mouth to an unconscious person.  
If symptoms persist, call a medical doctor.

General advice : Move out of dangerous area.  
Show this safety data sheet to the doctor in attendance.  
Do not leave the victim unattended.

Most important symptoms and effects, both acute and delayed : No information available.

### 5. FIREFIGHTING MEASURES

Unsuitable extinguishing media : High volume water jet

Specific extinguishing methods : Standard procedure for chemical fires.  
Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Special protective equipment for firefighters : Wear self-contained breathing apparatus for firefighting if necessary.

## 6. ACCIDENTAL RELEASE MEASURES

- Personal precautions, protective equipment and emergency procedures : Use personal protective equipment.
- Environmental precautions : Prevent product from entering drains.  
Prevent further leakage or spillage if safe to do so.
- Methods and materials for containment and cleaning up : Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust).  
Keep in suitable, closed containers for disposal.

## 7. HANDLING AND STORAGE

- Advice on protection against fire and explosion : Normal measures for preventive fire protection.
- Advice on safe handling : Do not breathe vapours/dust.  
Avoid exposure - obtain special instructions before use.  
For personal protection see section 8.  
Smoking, eating and drinking should be prohibited in the application area.  
Dispose of rinse water in accordance with local and national regulations.  
Persons susceptible to skin sensitisation problems or asthma, allergies, chronic or recurrent respiratory disease should not be employed in any process in which this mixture is being used.  
Avoid contact with skin and eyes.
- Conditions for safe storage : Keep container tightly closed in a dry and well-ventilated place.  
Electrical installations / working materials must comply with the technological safety standards.

## 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

### Components with workplace control parameters

| Components                                                                                      | CAS-No.    | Value type (Form of exposure) | Control parameters / Permissible concentration | Basis |
|-------------------------------------------------------------------------------------------------|------------|-------------------------------|------------------------------------------------|-------|
| Paraffinic Distillate, Heavy, Hydrotreated                                                      | 64742-54-7 | TWA (Inhalable fraction)      | 5 mg/m <sup>3</sup>                            | ACGIH |
| Further information: Upper Respiratory Tract irritation, Not classifiable as a human carcinogen |            |                               |                                                |       |

### Personal protective equipment

#### Hand protection

- Remarks : The suitability for a specific workplace should be discussed with the producers of the protective gloves.

#### Eye protection

- : Eye wash bottle with pure water.  
Tightly fitting safety goggles.  
Wear face-shield and protective suit for abnormal processing problems.

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|                          |                                                                                                                                          |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Skin and body protection | : Impervious clothing.<br>Choose body protection according to the amount and concentration of the dangerous substance at the work place. |
| Hygiene measures         | : When using do not eat or drink.<br>When using do not smoke.<br>Wash hands before breaks and at the end of workday.                     |

### 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                         |                                                                     |
|-----------------------------------------|---------------------------------------------------------------------|
| Appearance                              | : Viscous liquid                                                    |
| Initial boiling point and boiling range | : > 300 °C                                                          |
| Flash point                             | : 210 °C Method: ASTM D92, open cup                                 |
| Density                                 | : 0.885 g/cm <sup>3</sup> (15 °C)<br>Method: ASTM D 4052            |
| Viscosity, kinematic                    | : 9.5 mm <sup>2</sup> /s (100 °C)<br>114 mm <sup>2</sup> /s (40 °C) |

### 10. STABILITY AND REACTIVITY

|                                    |                                                               |
|------------------------------------|---------------------------------------------------------------|
| Reactivity                         | : Hazardous polymerisation does not occur.                    |
| Chemical stability                 | : Stable under normal conditions.                             |
| Possibility of hazardous reactions | : No dangerous reaction known under conditions of normal use. |
| Conditions to avoid                | : Heat, sparks, flame and build-up of static electricity.     |
| Incompatible materials             | : Strong oxidizing agents.                                    |
| Hazardous decomposition products   | : Fumes, smoke, carbon monoxide                               |

### 11. TOXICOLOGICAL INFORMATION

#### Acute toxicity

##### Components:

##### **Paraffinic Distillate, Heavy, Hydrotreated:**

|                           |                                |
|---------------------------|--------------------------------|
| Acute oral toxicity       | : LD50 (Rat): > 5,000 mg/kg    |
| Acute inhalation toxicity | : LC50 (Rat): > 5.53 mg/l      |
| Acute dermal toxicity     | : LD50 (Rabbit): > 2,000 mg/kg |

##### **Benzenesulfonic acid, propenated, calcium salt, overbased:**

|                           |                                              |
|---------------------------|----------------------------------------------|
| Acute oral toxicity       | : LD50 (Rat, Male and female): > 5,000 mg/kg |
| Acute inhalation toxicity | : Remarks: No data available                 |
| Acute dermal toxicity     | : LD50 (Rat): > 5,000 mg/kg                  |

#### Skin corrosion/irritation

##### Product:

|         |                                                |
|---------|------------------------------------------------|
| Remarks | : May cause skin irritation and/or dermatitis. |
|---------|------------------------------------------------|

##### Components:

##### **Paraffinic Distillate, Heavy, Hydrotreated:**

|         |                      |
|---------|----------------------|
| Species | : Rabbit             |
| Result  | : No skin irritation |

##### **Zinc Dialkyl Dithiophosphate:**

|        |                   |
|--------|-------------------|
| Result | : Skin irritation |
|--------|-------------------|

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### Benzenesulfonic acid, propenated, calcium salt, overbased:

Species : Rabbit  
Exposure time : 4 h  
Result : No skin irritation

### Serious eye damage/eye irritation

#### Product:

Remarks : Vapours may cause irritation to the eyes, respiratory system and the skin.

#### Components:

##### Paraffinic Distillate, Heavy, Hydrotreated:

Species : Rabbit  
Result : No eye irritation

##### Zinc Dialkyl Dithiophosphate:

Result : Irreversible effects on the eye

### Benzenesulfonic acid, propenated, calcium salt, overbased:

Remarks : No data available

### Respiratory or skin sensitisation

#### Components:

##### Paraffinic Distillate, Heavy, Hydrotreated:

Exposure routes : Skin contact  
Result : No evidence of skin sensitization.  
Exposure routes : Inhalation  
Remarks : No data available

##### Benzenesulfonic acid, propenated, calcium salt, overbased:

Species : Guinea pig  
Result : May cause sensitisation by skin contact.

### Germ cell mutagenicity

#### Components:

##### Paraffinic Distillate, Heavy, Hydrotreated:

Germ cell mutagenicity - : No data available  
Assessment

##### Benzenesulfonic acid, propenated, calcium salt, overbased:

Germ cell mutagenicity - : No data available  
Assessment

### Carcinogenicity

#### Components:

##### Paraffinic Distillate, Heavy, Hydrotreated:

Carcinogenicity - Assessment : Contains less than 3 % DMSO extract as measured by IP 346  
'Determination of polycyclic aromatics in unused lubricating base oils and asphaltene free petroleum fractions — Dimethyl sulphoxide extraction refractive index method'

##### Benzenesulfonic acid, propenated, calcium salt, overbased:

Carcinogenicity - Assessment : No data available

### Reproductive toxicity

#### Components:

##### Paraffinic Distillate, Heavy, Hydrotreated:

Reproductive toxicity - : Not suspected to be a reproductive toxicant.  
Assessment

**Benzenesulfonic acid, propenated, calcium salt, overbased:**

Reproductive toxicity - : No data available  
Assessment

**STOT - single exposure**

**Components:**

**Paraffinic Distillate, Heavy, Hydrotreated:**

Assessment : No specific organ toxicity following single exposure.

**Benzenesulfonic acid, propenated, calcium salt, overbased:**

Remarks : No data available

**STOT - repeated exposure**

**Components:**

**Paraffinic Distillate, Heavy, Hydrotreated:**

Assessment : No specific organ toxicity following repeated exposure.

**Benzenesulfonic acid, propenated, calcium salt, overbased:**

Remarks : No data available

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**12. ECOLOGICAL INFORMATION**

**Ecotoxicity**

**Components:**

**Paraffinic Distillate, Heavy, Hydrotreated:**

Toxicity to fish : LL50 (Pimephales promelas (fathead minnow)): > 100 mg/l  
Exposure time: 96 h

Toxicity to daphnia and other aquatic invertebrates : Remarks: No data available

Toxicity to algae : NOEL (Pseudokirchneriella subcapitata (green algae)): >= 100 mg/l  
Exposure time: 72 h

Toxicity to fish (Chronic toxicity) : NOEL (Oncorhynchus mykiss (rainbow trout)): >= 1,000 mg/l  
Exposure time: 14 d

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEL (Daphnia magna (Water flea)): 10 mg/l  
Exposure time: 21 d

Toxicity to microorganisms : NOEL: > 1.93 mg/l  
Exposure time: 4 d

**Zinc Dialkyl Dithiophosphate:**

**Ecotoxicology Assessment**

Acute aquatic toxicity : Toxic to aquatic life.

Chronic aquatic toxicity : Toxic to aquatic life with long lasting effects.

**HOB Calcium Long Chain Alkyl Sulfonate:**

**Ecotoxicology Assessment**

Chronic aquatic toxicity : May cause long lasting harmful effects to aquatic life.

**Benzenesulfonic acid, propenated, calcium salt, overbased:**

Toxicity to fish : LL50 (Oncorhynchus mykiss (rainbow trout)): > 100 mg/l  
Exposure time: 96 h

Toxicity to daphnia and other aquatic invertebrates : EL50 (Daphnia magna (Water flea)): > 1,000 mg/l  
Exposure time: 48 h

Toxicity to algae : EL50 (Pseudokirchneriella subcapitata (algae)): > 1,000 mg/l

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Exposure time: 72 h  
Toxicity to fish : Remarks: No data available  
(Chronic toxicity)  
Toxicity to daphnia and other : Remarks: No data available  
aquatic invertebrates  
(Chronic toxicity)  
Toxicity to microorganisms : Remarks: No data available  
Ecotoxicology Assessment  
Chronic aquatic toxicity : May cause long lasting harmful effects to aquatic life.

### Persistence and degradability

#### Components:

##### **Paraffinic Distillate, Heavy, Hydrotreated:**

Biodegradability : Exposure time: 28 d  
Result: Not readily biodegradable. Method: OECD Test  
Guideline 301

##### **Benzenesulfonic acid, propenated, calcium salt, overbased:**

Biodegradability : Result: Not readily biodegradable.

### Bioaccumulative potential

#### Components:

##### **Paraffinic Distillate, Heavy, Hydrotreated:**

Bioaccumulation : Remarks: No data available  
Partition coefficient: : Remarks: No data available  
n-octanol/water

##### **Benzenesulfonic acid, propenated, calcium salt, overbased:**

Bioaccumulation : Concentration: 0.01 mg/l  
Bioconcentration factor (BCF): 58  
Remarks: Potentially bio-accumulative.  
Partition coefficient: : log Pow: 5.70  
n-octanol/water

### Mobility in soil

#### Components:

##### **Paraffinic Distillate, Heavy, Hydrotreated:**

Mobility : Medium: Soil  
Remarks: No data available

### Other adverse effects

#### Product:

Additional ecological : No data available  
information

#### Components:

##### **Paraffinic Distillate, Heavy, Hydrotreated:**

Additional ecological : No data available  
information

**Zinc Dialkyl Dithiophosphate:**No data available  
**HOB Calcium Long Chain Alkryl Sulfonate:**No data available  
**Benzenesulfonic acid, propenated, calcium salt, overbased:**No data available

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### 13. DISPOSAL CONSIDERATIONS

#### Disposal methods

- |                        |   |                                                                                                                                                                        |
|------------------------|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Waste from residues    | : | Do not dispose of waste into sewer.<br>Do not contaminate ponds, waterways or ditches with chemical or used container.<br>Send to a licensed waste management company. |
| Contaminated packaging | : | Empty remaining contents.<br>Dispose of as unused product.<br>Do not re-use empty containers.                                                                          |

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### 14. TRANSPORT INFORMATION

#### International Regulation

##### UNRTDG

Not regulated as a dangerous good

##### IATA-DGR

Not regulated as a dangerous good

##### IMDG-Code

Not regulated as a dangerous good

#### Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

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### 15. REGULATORY INFORMATION

#### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### The components of this product are reported in the following inventories:

- |        |   |                                                                                                                                                                                                                                                                     |
|--------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CH INV | : | Not in compliance with the inventory                                                                                                                                                                                                                                |
| TSCA   | : | Not On TSCA Inventory                                                                                                                                                                                                                                               |
| DSL    | : | This product contains the following components that are not on the Canadian DSL nor NDSL.<br>Mineral Oil<br>Highly refined mineral oil (C15 - C50)<br>Acrylic copolymer<br>HOB Calcium Long Chain Alkryl Sulfonate<br>Interchangeable neutral oils<br>Aryl diesters |
| AICS   | : | Not in compliance with the inventory                                                                                                                                                                                                                                |
| NZIoC  | : | Not in compliance with the inventory                                                                                                                                                                                                                                |
| ENCS   | : | Not in compliance with the inventory                                                                                                                                                                                                                                |
| ISHL   | : | Not in compliance with the inventory                                                                                                                                                                                                                                |
| KECI   | : | Not in compliance with the inventory                                                                                                                                                                                                                                |
| PICCS  | : | Not in compliance with the inventory                                                                                                                                                                                                                                |
| IECSC  | : | Not in compliance with the inventory                                                                                                                                                                                                                                |



## 16. OTHER INFORMATION

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### Full text of other abbreviations

|         |   |                                                                                                     |
|---------|---|-----------------------------------------------------------------------------------------------------|
| (Q)SAR  | - | (Quantitative) Structure Activity Relationship                                                      |
| ACGIH   | - | American Conference of Governmental Industrial Hygienists                                           |
| AICS    | - | Australian Inventory of Chemical Substances                                                         |
| ANTT    | - | National Agency for Transport by Land of Brazil                                                     |
| ASTM    | - | American Society for the Testing of Materials                                                       |
| bw      | - | Body weight                                                                                         |
| CCHC    | - | Chemicals Classification and Hazard Communication                                                   |
| CMR     | - | Carcinogen, Mutagen or Reproductive Toxicant                                                        |
| CPR     | - | Controlled Products Regulations                                                                     |
| DIN     | - | Standard of the German Institute for Standardisation                                                |
| DSL     | - | Domestic Substances List (Canada)                                                                   |
| ECx     | - | Concentration associated with x% response                                                           |
| ELx     | - | Loading rate associated with x% response                                                            |
| EmS     | - | Emergency Schedule                                                                                  |
| ENCS    | - | Existing and New Chemical Substances (Japan)                                                        |
| ErCx    | - | Concentration associated with x% growth rate response                                               |
| ERG     | - | Emergency Response Guide                                                                            |
| GHS     | - | Globally Harmonized System                                                                          |
| GLP     | - | Good Laboratory Practice                                                                            |
| IARC    | - | International Agency for Research on Cancer                                                         |
| IATA    | - | International Air Transport Association                                                             |
| IBC     | - | International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk |
| IC50    | - | Half maximal inhibitory concentration                                                               |
| ICAO    | - | International Civil Aviation Organization                                                           |
| ICOP    | - | Industry Code of Practice on Chemicals Classification and Hazard Communication                      |
| IECSC   | - | Inventory of Existing Chemical Substances in China                                                  |
| IMDG    | - | International Maritime Dangerous Goods                                                              |
| IMO     | - | International Maritime Organization                                                                 |
| ISHL    | - | Industrial Safety and Health Law (Japan)                                                            |
| ISO     | - | International Organisation for Standardization                                                      |
| KECI    | - | Korea Existing Chemicals Inventory                                                                  |
| LC50    | - | Lethal Concentration to 50 % of a test population                                                   |
| LD50    | - | Lethal Dose to 50% of a test population (Median Lethal Dose)                                        |
| MARPOL  | - | International Convention for the Prevention of Pollution from Ships                                 |
| MY PEL  | - | Malaysian Permissible Exposure Limit                                                                |
| n.o.s.  | - | Not Otherwise Specified                                                                             |
| Nch     | - | Chilean Norm                                                                                        |
| NITE    | - | National Institute of Technology and Evaluation                                                     |
| NO(A)EC | - | No Observed (Adverse) Effect Concentration                                                          |
| NO(A)EL | - | No Observed (Adverse) Effect Level                                                                  |
| NOELR   | - | No Observable Effect Loading Rate                                                                   |
| NOM     | - | Official Mexican Norm                                                                               |
| NTP     | - | National Toxicology Program                                                                         |
| NZIoC   | - | New Zealand Inventory of Chemicals                                                                  |
| OCSP    | - | Office of Chemical Safety and Pollution Prevention                                                  |
| OECD    | - | Organization for Economic Co-operation and Development                                              |
| PBT     | - | Persistent, Bioaccumulative and Toxic                                                               |
| PICCS   | - | Philippines Inventory of Chemicals and Chemical Substances                                          |
| REACH   | - | Regulation (EC) No 1907/2006 of the European Parliament and of the Council                          |

|        |   |                                                                                     |
|--------|---|-------------------------------------------------------------------------------------|
|        |   | concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals |
| SADT   | - | Self-Accelerating Decomposition Temperature                                         |
| SDS    | - | Safety Data Sheet                                                                   |
| STEL   | - | Short Term Exposure Limit                                                           |
| TCSI   | - | Taiwan Chemical Substance Inventory                                                 |
| TDG    | - | Transportation of Dangerous Goods                                                   |
| TSCA   | - | Toxic Substances Control Act (United States)                                        |
| TWA    | - | Time Weighted Average                                                               |
| UN     | - | United Nations                                                                      |
| UNRTDG | - | United Nations Recommendations on the Transport of Dangerous Goods                  |
| UVCB   | - | Unknown or Variable Composition, Complex Reaction Products and Biological Materials |
| vPvB   | - | Very Persistent and Very Bioaccumulative                                            |
| WHMIS  | - | Workplace Hazardous Materials Information System                                    |

#### Disclaimer

To the best of PETRONAS knowledge, the information contained herein is accurate and reliable as of the date of publication. Any use of information provided should be undertaken in consultation with a professional with the appropriate expertise, as necessary in the circumstances. No warranty, express or implied, is given as to the quality, accuracy, reliability, applicability or completeness of the contents of this SDS. The information presented here pertains only to the product as shipped. It is the responsibility of the customer to ensure that any activities relating to the product comply with all federal, state or local laws. Any hazards associated with any product regulatory requirements are subject to change and may differ between various locations. Except to the extent required by law, re-publication or retransmission of this SDS, in whole or in part, is strictly prohibited. PETRONAS does not take responsibility for use, transportation, storage, handling, packaging or disposal of the product that is beyond our knowledge and recommendation by this SDS.

#### Product Stewardship Advisory:

PETRONAS aims to increase awareness of all the hazards associated with the storage, handling and use of its products. Thoroughly reviewing the accompanying Safety Data Sheets and disseminating the information to all dependent and interested parties is an essential part of any 'Responsible Care' programme.

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